



# RAMCO INSTITUTE OF TECHNOLOGY

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## Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8691 & Artificial Intelligence

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/CSE

### Innovative Practice Description

- **Unit / Topic:** Unit II / Local Search Algorithms
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** Crossword Puzzle
- **Justification:**

Local Search algorithm and optimization problems are the significant search strategies in Artificial Intelligence. Different types of search strategies are available to solve the problem, we should select the best way to solve the problem and obtain the optimal solution. Hill climbing algorithm is a local search algorithm which continuously moves in the direction of increasing elevation/value to find the peak of the mountain or best solution to the problem. It terminates when it reaches a peak value where no neighbor has a higher value. Various local search algorithms are easily paired with problems with the help of crossword puzzle which leads to the greater clarity about the concept of LSA to the students.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
  - A crossword puzzle is a fun word game that encourages students to remember the idea and try to figure out the solution on their own. Instead of administering standard MCQ tests, it sparks pupils' curiosity.
  - The activity was conducted in a classroom for a duration of 10 minutes. The students were given 10 questions as a crossword puzzle in the topic of Local Search Algorithm. They have to think and write the answers in the given puzzle.
  - The activity was conducted during the last 10 minutes of the class after that the answers were corrected. Sample answers were shown in figure 1. The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.
- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO9 | PO10 | PO12 | PSO1 |
|-----|-----|-----|-----|-----|------|------|------|
| CO2 | 3   | 2   | 1   | 1   | 2    | 1    | 2    |

(1 – Low      2 – Moderate      3 – High)

• PO / PSO mapped:

| Innovative practice           | PO1                                          | PO2                                      | PO3                                                     | PO9                                                                   | PO10                                                                                                                        | PO12                                                                                           | PSO1                                                               |
|-------------------------------|----------------------------------------------|------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|                               | 3                                            | 2                                        | 1                                                       | 1                                                                     | 2                                                                                                                           | 1                                                                                              | 1                                                                  |
| Justification for correlation | The students apply the knowledge of pointers | The students analyze the given questions | The students find the solutions for the given question. | The students work in a team to find solutions for the given question. | The students communicate with their team members and explain the reason for providing particular solution for the question. | Recognize the need for Local search Algorithm in the broadest context of technological change. | The students apply the knowledge of pointer to solve the question. |

• Images / Screenshot of the practice:

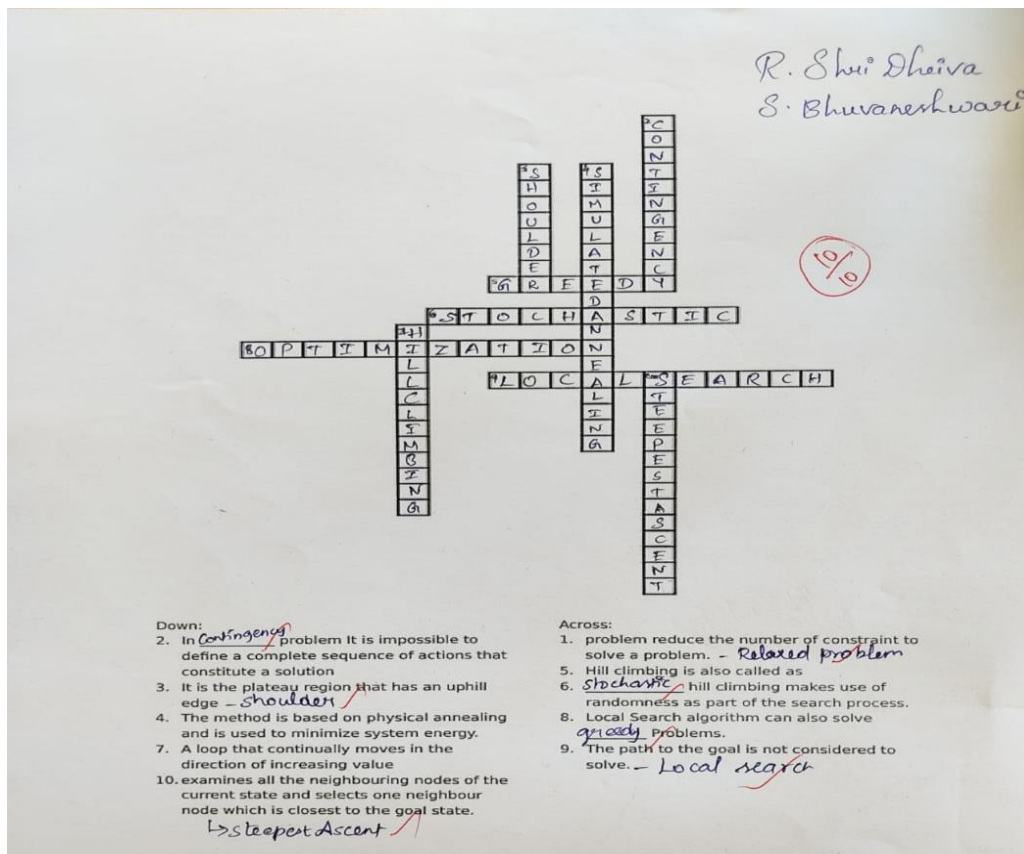
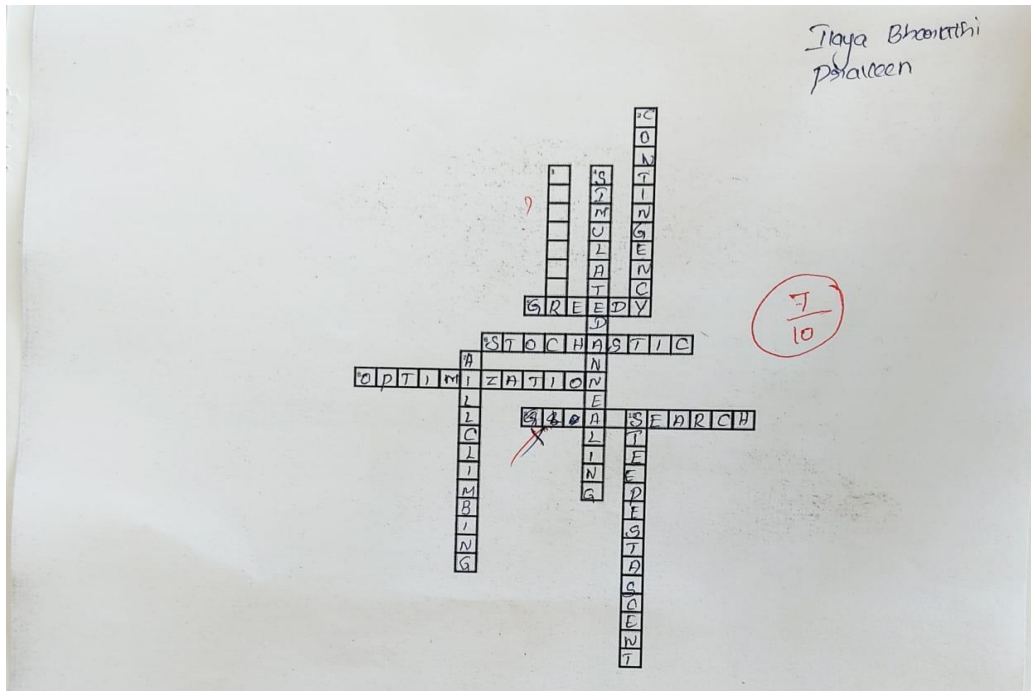


Fig 1: Sample Answers



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Some of the students find very easy to identify the answer for the given questions.
- Some of them take more time to complete the puzzle need more clarification on the various classes.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to think on their own and identify the use of local search algorithm for the variables and access the value of the variable by passing the address of the variable. The students level of understanding on this topic gets increased.

- ❖ ***Challenges faced in implementation:***

- The students who didn't understand the concept clearly were unable to complete it within the allotted time.
- Few of them were struggled to find the answer for the questions.

**References:**

- ❖ [https://www.ritrjpm.ac.in/images/computer-science/RV\\_crossword\\_BSEC.pdf](https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf)
- ❖ <https://crosswordlabs.com/>

**Signature of Faculty Member**

**HOD**